Asymmetric Generalization

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Generative model

The overall generative model can be thought of as a latent function ϕ that maps from the embedding space $\mathcal{X} = \mathbb{R}^n$ to a potential $\mathbb{R}$. In the simplest case, a learner observes (x, y) and needs to generalize to a new point x'. This is a posterior predictive:

$$P(\phi(x') \mid \phi(x) = y) = \int_{\Phi} P(\phi(x') \mid \phi) P(\phi \mid \phi(x) = y) \, \mathrm{d}\phi$$

where we have a posterior over the function:

$$P(\phi \mid \phi(x) = y) \propto P(\phi(x) = y \mid \phi) P(\phi)$$

with prior $P(phi.alt)$`{=typst} over functions.

GP formulation

In Gaussian Process (GP) terms, we can think of ϕ as a function sampled from a GP with mean function $m(x)$ and kernel $k(x, x')$.

$$\phi \sim \mathcal{GP}(m, k)$$

The standard RBF kernel with amplitude σ and lengthscale l is:

$$k_{\sigma,l}(x, x') = \sigma^2 \exp\left(-\frac{(x - x')^2}{2l^2}\right)$$

From this perspective, $P(\phi)$ is a prior distribution parameterized by m, σ, and l, which we as the scientists need to infer but are fixed for the learner. Figure 1 depicts this as a graphical model.

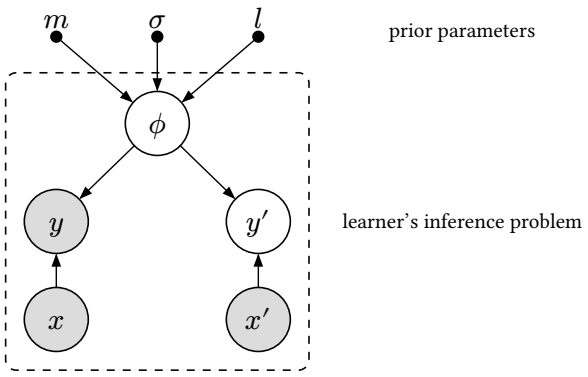


Figure 1: The learner's inference problem. Shaded nodes are observed. Dots are prior parameters, fixed for the learner and inferred by the scientist.

Symmetric Generalization

Under the standard RBF, generalization is always symmetric, meaning the effect of observing (x, y) on a prediction at x' is the same as the effect of observing (x', y) on a prediction at x. Formally, given $x, x' \in \mathcal{X}$ and $y \in \mathbb{R}$ generalization is symmetric (assuming a constant mean function):

$$P(\phi(x) \mid \phi(x') = y) = P(\phi(x') \mid \phi(x) = y)$$

This is because the covariance matrix induced by the standard RBF is symmetric (as it must be) and its diagonal is constant. For example, take the unit RBF $k = k_{1,1}$, with $x = -1$, $x' = 1$, and $y = 1$. The covariance matrix is:

$$K = \begin{pmatrix} k(-1,-1) & k(-1,1) \\ k(1,-1) & k(1,1) \end{pmatrix} \approx \begin{pmatrix} 1 & 0.135 \\ 0.135 & 1 \end{pmatrix}$$

With zero mean, the two conditional distributions have the same mean and variance:

$$\phi(1) \mid \phi(-1) = 1 \sim \mathcal{N}(0.135, 0.982)$$

$$\phi(-1) \mid \phi(1) = 1 \sim \mathcal{N}(0.135, 0.982)$$

Asymmetric Generalization with Non-stationary Amplitude

If the covariance matrix can have non-constant diagonal elements, then generalization can be asymmetric. For example, scale the unit RBF k by an amplitude $\sigma(x)$ that differs between the two points, $\sigma(-1) = 1.414$ and $\sigma(1) = 0.707$. The covariance matrix is then:

$$K = \begin{pmatrix} \sigma(-1)^2 k(-1,-1) & \sigma(-1)\sigma(1)k(-1,1) \\ \sigma(1)\sigma(-1)k(1,-1) & \sigma(1)^2 k(1,1) \end{pmatrix} \approx \begin{pmatrix} 2 & 0.135 \\ 0.135 & 0.5 \end{pmatrix}$$

With the same $x = -1$, $x' = 1$, $y = 1$, and zero mean, the two conditional distributions differ:

$$\phi(1) \mid \phi(-1) = 1 \sim \mathcal{N}(0.068, 0.491)$$

$$\phi(-1) \mid \phi(1) = 1 \sim \mathcal{N}(0.270, 1.964)$$

The correlation between the two points is the same as before, 0.135. But observing at $x' = 1$, where the prior variance is smaller, moves the prediction at $x = -1$ four times as much as the reverse.

Therefore, we can define an RBF kernel with a non-stationary amplitude to achieve asymmetric generalization. The amplitude becomes a real-valued function of the input, $\sigma : \mathcal{X} \to \mathbb{R}$:

$$k_{\sigma(\cdot),l}(x, x') = (\sigma(x)\sigma(x') + \varepsilon^2) \exp\left(-\frac{(x - x')^2}{2l^2}\right)$$

where $\varepsilon \geq 0$ is a small constant. With $\varepsilon = 0$ the kernel scales the RBF by $\sigma(x)\sigma(x')$, and the covariance matrix in the example above is the one induced by $k_{\sigma(\cdot),1}$. With $\varepsilon > 0$ the prior variance never reaches zero, even where the amplitude does.

Essentialness

Amplitude roughly corresponds to the intrinsic variability of a point. Where the amplitude is large in magnitude (high variance), a given observed value is unremarkable and generalizes less. Where it is small (low variance), the same value is a large departure from the prior and generalizes more. We can then define a measure of "essentialness" as the inverse of the magnitude of the amplitude:

$$e(x) = \frac{1}{\sqrt{\sigma(x)^2 + \varepsilon^2}}$$

where ε is a small constant that ensures essentialness is bounded.

Single observation example

To illustrate, we use the kernel $k_{\sigma(\cdot),1}$ with $\varepsilon = 0.1$ and zero prior mean. We hold the amplitude at $x = -1$ at 1 and compare three values of the amplitude at $x' = 1$: 0.25, 1, and 4. These give an essentialness at $x' = 1$ of about 3.7, 1, and 0.25. In between, the amplitude varies smoothly. With $\sigma(1) = 1$ the amplitude is constant, so the kernel is a standard RBF.

> **i** How the amplitude function is chosen
>
> The amplitude is a smooth real-valued curve through $\sigma(-1) = 1$ and the chosen $\sigma(1)$. It is computed as the posterior mean of a GP with prior mean 1 and the unit RBF kernel. In data analysis σ is latent and has to be inferred. Here we fix it by hand.

Figure 2 has one column per amplitude function. The top row shows the amplitude function, in red. The middle row shows the essentialness, in purple. The bottom row shows the posterior mean of ϕ, in blue, for two separate observations: $y = 1$ at $x = -1$, or $y = 1$ at $x' = 1$. Each curve passes through its observation (filled circle), and the open circle marks its prediction at the other point.

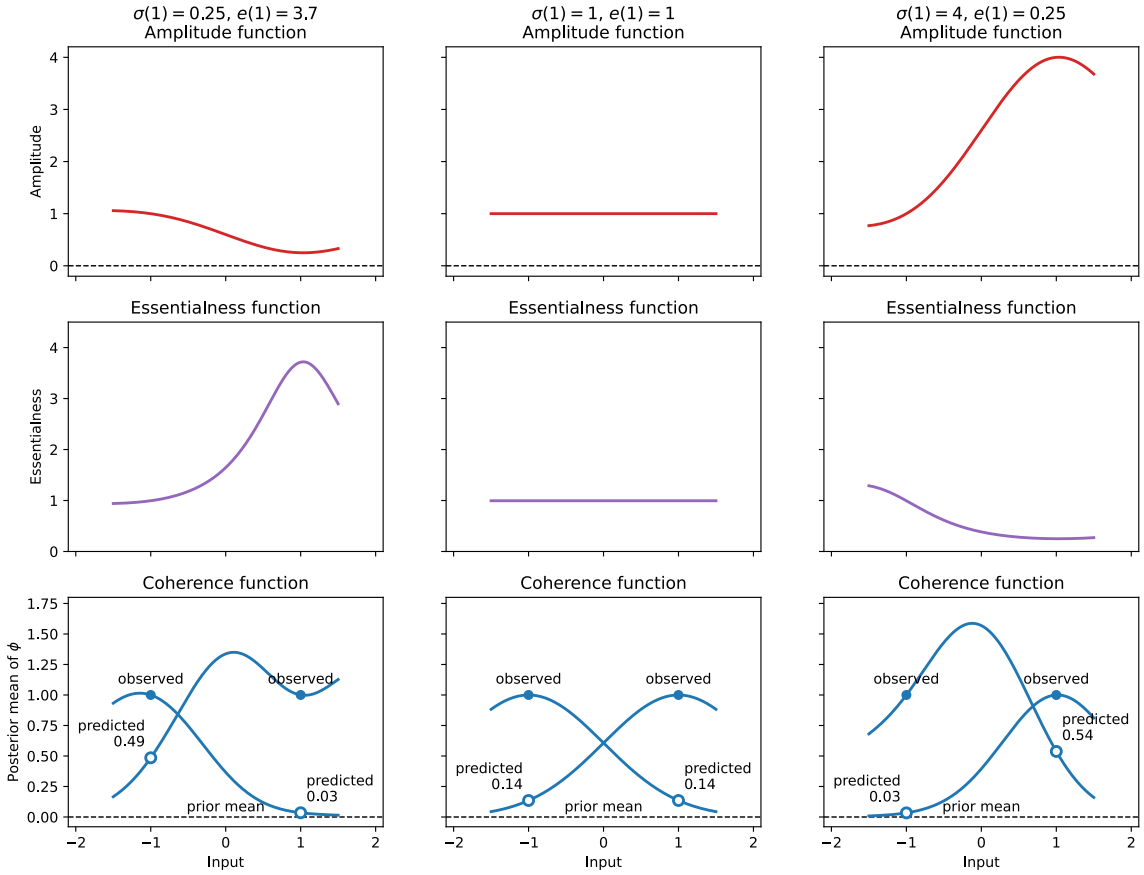


Figure 2: Amplitude functions (top), essentialness functions (middle), and posterior means of ϕ (bottom) for three values of $\sigma(1)$. Filled circles mark observations and open circles mark predictions at the other point.

With constant amplitude, $\sigma(1) = 1$, both predictions are 0.14. With $\sigma(1) = 0.25$, the point $x' = 1$ is about 3.7 times as essential as $x = -1$. Observing at $x' = 1$ predicts 0.49 at $x = -1$, while observing at $x = -1$ predicts only 0.03 at $x' = 1$. With $\sigma(1) = 4$ the pattern reverses, with predictions of 0.54 and 0.03.

Multiple observation example

Figure 3 shows an example of multiple observations under different amplitude and essentialness functions.

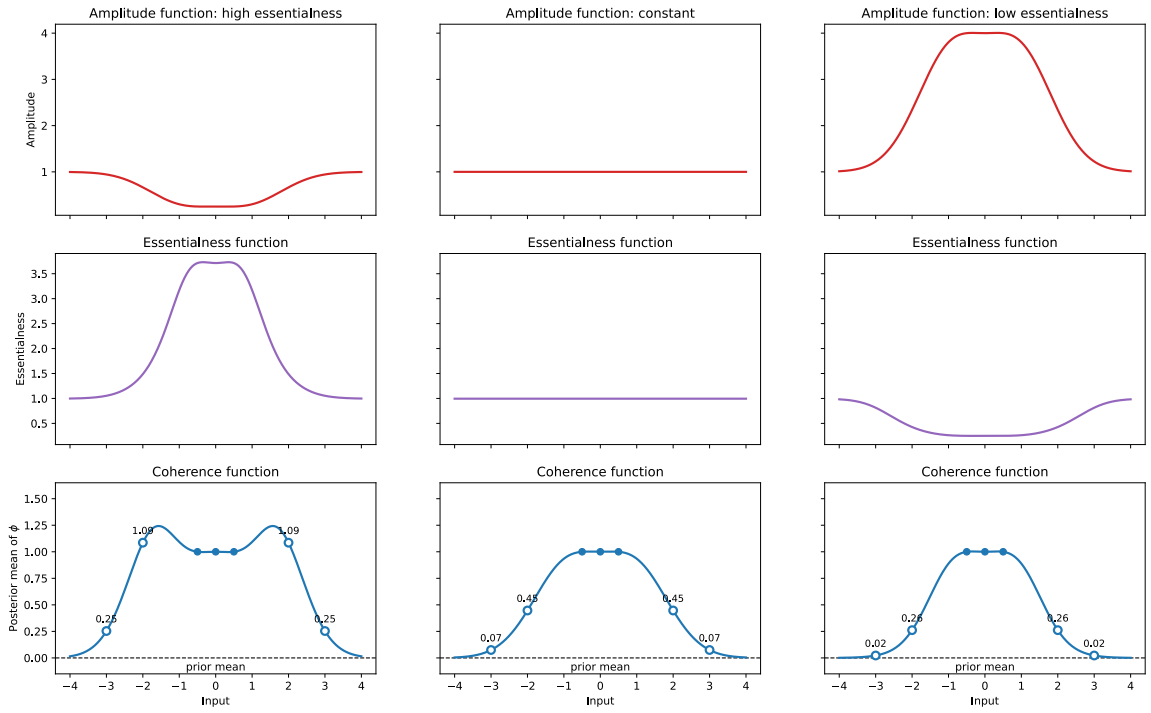


Figure 3: Amplitude functions (top), essentialness functions (middle), and posterior means of ϕ (bottom) when a cluster of three observations has high, constant, or low essentialness. Filled circles mark the observations and open circles mark predictions further away.

Asymmetric Generalization with Non-stationary Lengthscale

Another way to make the RBF kernel non-stationary is to let the lengthscale depend on the input, $l : \mathcal{X} \to \mathbb{R}_{>0}$. This is the Gibbs kernel:

$$k_{\sigma,l(\cdot)}(x, x') = \sigma^2 \sqrt{\frac{2l(x)l(x')}{l(x)^2 + l(x')^2}} \exp\left(-\frac{(x - x')^2}{l(x)^2 + l(x')^2}\right)$$

When $l(x) = c$ is constant, this is the standard RBF $k_{\sigma,c}$.

Single observation example

Unlike a non-stationary amplitude, a non-stationary lengthscale does not make generalization between two points asymmetric. The diagonal of the covariance matrix stays constant, since $k_{\sigma,l(\cdot)}(x, x) = \sigma^2$ for every x. For example, take $\sigma = 1$, $l(-1) = 1$, and $l(1) = 0.25$. With $x = -1$, $x' = 1$, and $y = 1$, the covariance matrix is:

$$K \approx \begin{pmatrix} 1 & 0.016 \\ 0.016 & 1 \end{pmatrix}$$

and the two conditional distributions have the same mean and variance:

$$\phi(1) \mid \phi(-1) = 1 \sim \mathcal{N}(0.016, 1.000)$$

$$\phi(-1) \mid \phi(1) = 1 \sim \mathcal{N}(0.016, 1.000)$$

Figure 4 repeats the single observation example with the lengthscale in place of the amplitude. We hold the lengthscale at $x = -1$ at 1 and compare three values of the lengthscale at $x' = 1$: 0.25, 1, and 4. The log lengthscale is a smooth curve through these two values, computed as the posterior mean of a GP with the unit RBF kernel.

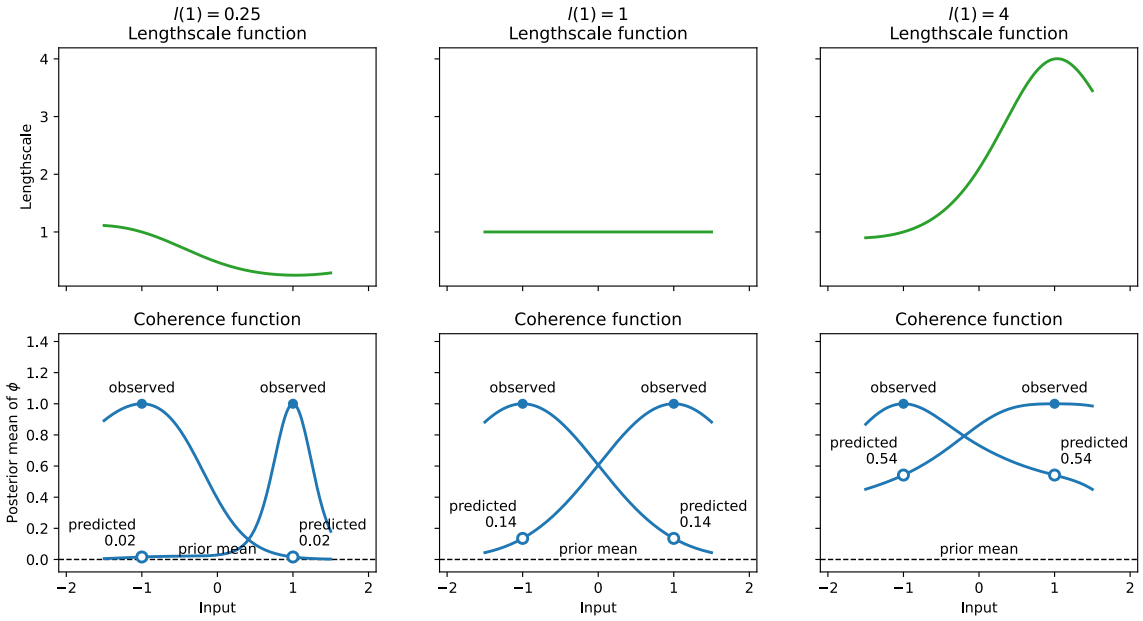


Figure 4: Lengthscale functions (top) and posterior means of ϕ (bottom) for three values of $l(1)$. Filled circles mark observations and open circles mark predictions at the other point.

The lengthscale changes how strongly an observation generalizes. The prediction at the other point is 0.02 with $l(1) = 0.25$, 0.14 with $l(1) = 1$, and 0.54 with $l(1) = 4$. But in every column the two predictions are equal, even though the two posterior means are no longer mirror images. A non-stationary lengthscale changes the strength of generalization between two points equally in both directions. Asymmetry between two points requires a non-constant diagonal, which the amplitude provides.

Multiple observation example

The symmetry above is between two single points. With several observations per side, a non-stationary lengthscale can make generalization asymmetric, by a different route than the amplitude.

Figure 5 shows how this happens. We observe $y = 1$ at both points on one side and predict the two points on the other side. With constant lengthscale the two posterior means are mirror images, and each set predicts 0.46 at the nearer point of the other set. With varying lengthscale, the observations on the left predict 0.74 at $x = 0.5$, while the observations on the right predict only 0.40 at $x = -0.5$.

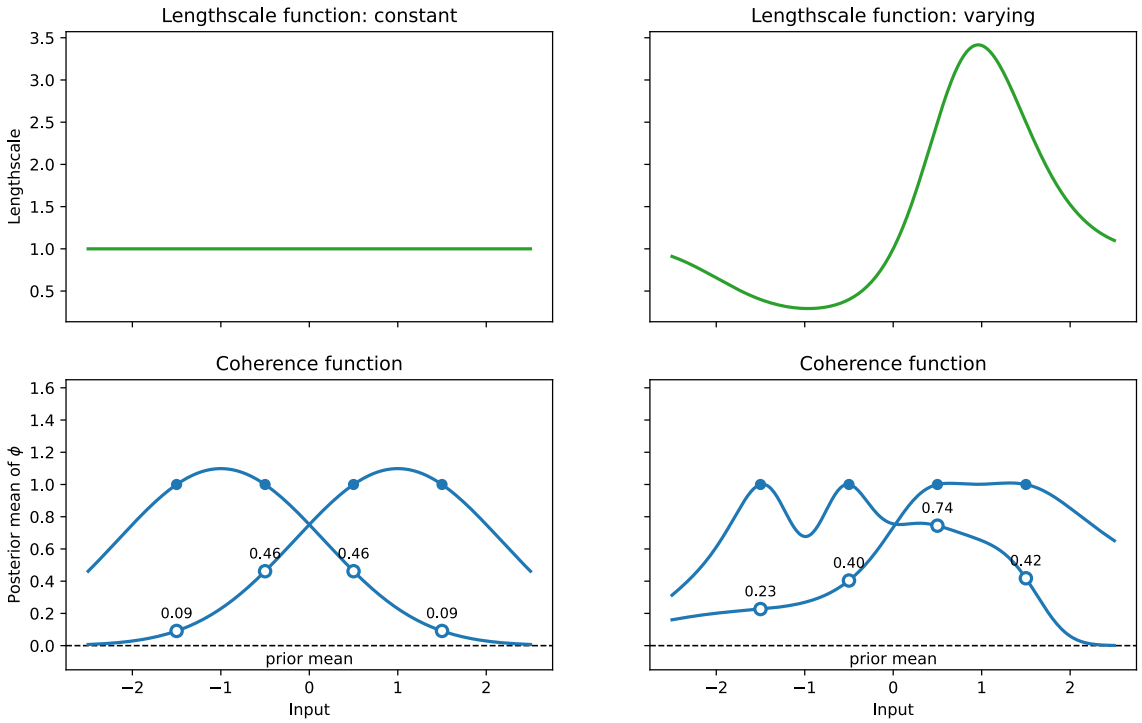


Figure 5: Lengthscale functions (top) and posterior means of ϕ (bottom) for a constant and a varying lengthscale. Each curve is conditioned on two observations (filled circles). Open circles mark its predictions at the inputs of the other pair.

The reason is redundancy. Where the lengthscale is short, the two observations are nearly independent, so they count as two pieces of evidence and their effects add. Where the lengthscale is long, the second observation mostly repeats the first, so together they count as little more than one.

This differs from the amplitude asymmetry in what it needs. A non-stationary amplitude makes generalization asymmetric even with one observation per side. A non-stationary lengthscale does so only with several observations per side.

Asymmetric Generalization with "Soft" Evidence

The examples in this document so far all condition on exact values of ϕ. But in our generics model, learners don't observe true values, but rather get generic statements that offer evidence about ϕ at the training features.

Because these are not exact observations, the learner's posterior belief about coherence y is influenced by two parts: the prior (defined by μ_0 and σ^2) and the likelihood of getting a generic given a speaker model.

But note that when σ is small (which is supposed to indicate high causal potential), our posterior for the coherence can't vary much from the prior mean, unless lots of data is observed. When σ is large, coherence moves closer to what the generic statement suggests more quickly.

Therefore, there's a tradeoff between flexibility of updating and causality when replacing an exact observation with an uncertain (Gaussian) one, $\tilde{y} = \phi(x) + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \tau^2)$. With zero prior mean, the prediction at x' is:

$$\mathbb{E}[\phi(x') \mid \tilde{y}] = \frac{\sigma(x')\sigma(x)\rho(x, x')}{\sigma(x)^2 + \tau^2}\tilde{y}$$

where $\rho(x, x')$ is the unit RBF correlation. With $\tau = 0$ this is the exact case, $\rho\frac{\sigma(x')}{\sigma(x)}\tilde{y}$, and the asymmetry is the ratio of amplitudes. With $\tau > 0$, a source with small amplitude moves very little itself, because its tight prior outweighs the evidence. In the limit $\sigma(x) \ll \tau$ the prediction becomes $\sigma(x')\sigma(x)\rho\frac{\tilde{y}}{\tau^2}$, which is **symmetric** in the two amplitudes. Soft evidence moves generalization from the exact case's amplitude ratio toward the amplitude product.

Multiple observations in a cluster

Consider two clusters on a line. Cluster A sits around $x = -1$, where the log-linear amplitude $\sigma(x) = e^{0.75x}$ is low, and has a test point at $x = -0.5$. Cluster B sits around $x = 1$, where the amplitude is high, with a test point at $x = 0.5$. Only cluster A receives evidence: $n = 2$, 3 or 4 soft observations $\tilde{y} = 1$ ($\tau = 1$), spread evenly over $[-1.3, -0.7]$. We then predict at both test points: A's own (A $\longrightarrow$ A) and B's (A $\longrightarrow$ B). We compare this amplitude with a constant one. The kernel is $k_{\sigma(\cdot),1}$ with $\varepsilon = 0$. This is the same log-linear form as the per-feature-prevalence model's non-stationary amplitude, which there uses the 2D feature embedding as x.

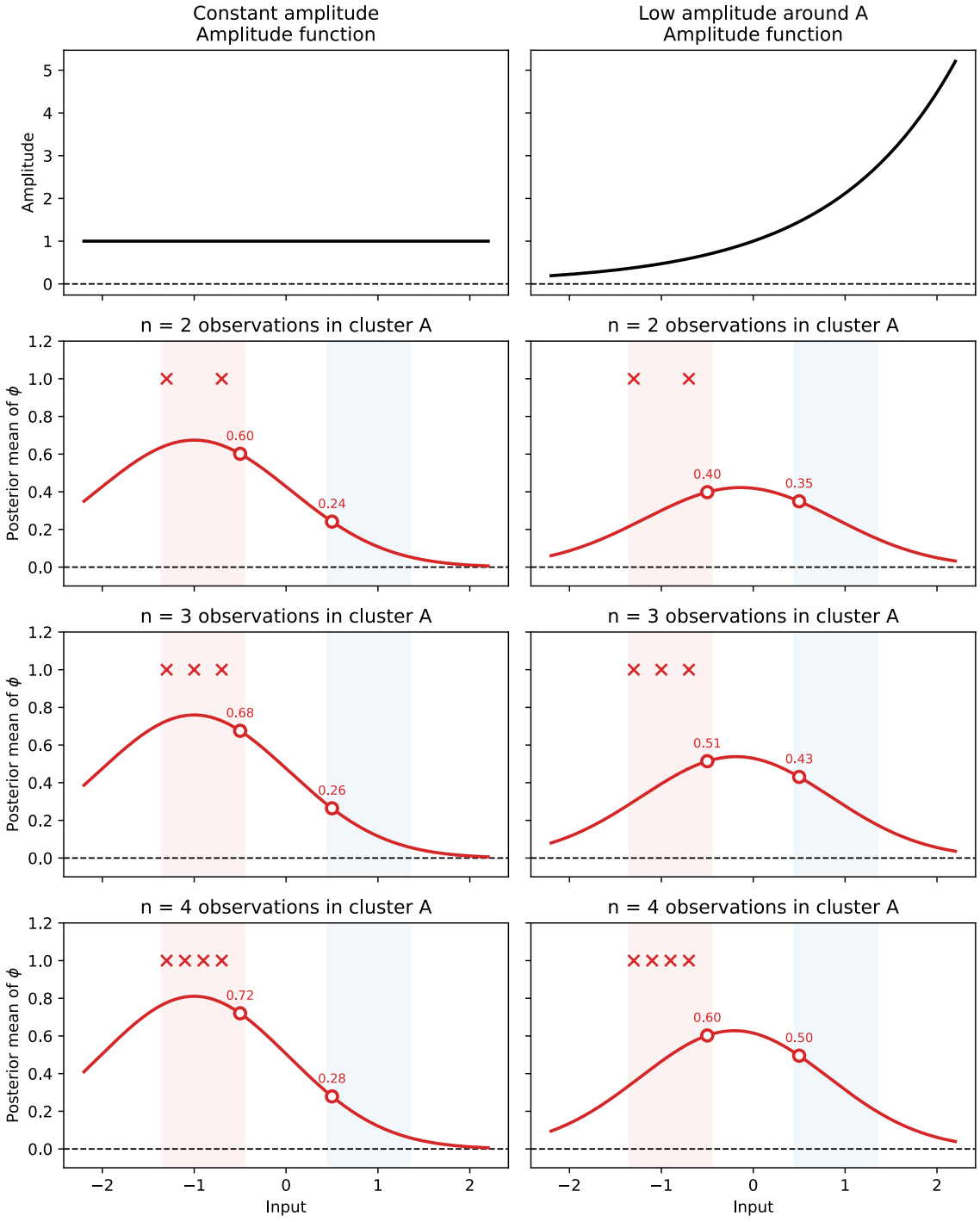


Figure 6: Amplitude functions (top) and posterior means of ϕ after $n = 2$, 3 or 4 soft observations in cluster A. Crosses mark the observations at their observed value and open circles mark predictions at the test points of cluster A (shaded red) and cluster B (shaded blue).

With one observation, the low amplitude around A keeps the posterior at A's testing point at 0.18, against 0.50 under a constant amplitude. A's own test point rises only 0.23, about half the constant-amplitude value (0.44). As the observations accumulate, two things happen:

1. **A's own boost recovers.** The evidence adds up against A's tight prior, so A's trained points reach 0.46 by $n = 4$. A's own boost rises to 0.60, close to the constant-amplitude value (0.72).
2. **A's evidence reaches B more and more.** Under a constant amplitude, B's test point gets about 40% of A's own boost, at every n. With the low amplitude around A, it gets 80–90%, because B's amplitude is high. B's absolute rise, though, depends on how far A's own points have moved: it is only 0.18 with one observation, and grows to 0.50 with four, against 0.28 under a constant amplitude.

So a low-amplitude cluster pays for its tight prior with a weak own boost when evidence is scarce. It becomes a strong source, one whose evidence reaches the high-amplitude region almost undiminished, only as evidence accumulates. The Gaussian observation in this example accumulates without limit. A generic does not. Its likelihood peaks at an intermediate coherence and nearby generics are largely redundant, so in the generics model the trade-off persists even with the 15 generics per condition of Study 9. In the Study 9 simulation, at the fitted parameters, the physical condition's own shift drops from 0.37 to 0.16 when the amplitude at physical is lowered by $|w| = 0.5$.

Implication for the generics model

The same structure holds exactly for the speaker likelihood. At the Laplace mode, $y - \mu_0 = K\alpha$, where α is the slope of $\log p(u \mid y)$ at each trained feature. So a test feature's shift is $\sum_s \sigma(x_t)\sigma(x_s)\rho(x_t, x_s)\alpha_s$, the amplitude product times the generic's remaining pull. The amplitude only creates asymmetry indirectly, through α: a low-amplitude source stays further below the generic's preferred coherence, so it keeps pulling harder.

In Study 9, physical generics raise physical features the most and also raise diet and personality features, while physical features barely move after diet or personality generics. A low amplitude at the physical features produces the third property and a little of the second, but it suppresses the first, as cluster A does in Figure 6 when evidence is scarce. A high amplitude does the reverse. So no amplitude gradient fits all three. Fitting the amplitude weights to Study 9 gives w near 0, and no improvement in the log marginal likelihood (log BF = −0.84). The directional pattern in the data needs a mechanism on the evidence side: generics about physical features have to carry more evidence or reach further, which acts on α rather than on the amplitude.